



DOSM Datathon 2026

“Tracking Tourist Trends for a Sustainable Malaysia ”

by

NULL

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1.Introduction

1.1 Background of study

Strong signs of recovery post COVID-19 have been shown by Malaysian domestic tourism. From data sourced from DOSM 2025, the number of domestic tourists stood at 290.1 million people with an annual average growth rate of 11.5%. Domestic tourism expenditure recorded RM121.3 billion. This demonstrates the relevance of domestic tourism as a major factor of economic development in Malaysia.

However, the growth has not been uniform. Domestic visitors are highly concentrated in certain states. For instance, in 2025 there are 36.4 million and 35.1 million domestic visitors visiting Selangor and W.P Kuala Lumpur respectively. These two states together had 25% of all domestic visitors in Malaysia. On the other hand, some states like Perlis, Wilayah Persekutuan Labuan and Wilayah Persekutuan Putrajaya have very few domestic visitors with only 3.66 million, 0.48 million and 2.71 million compared to the total visitors in Malaysia according to the DOSM 2025.

1.2 Problem Statement

There are two problems arising from this imbalance. First, it leads to over-tourism that is indicated by environmental degradation, strain on infrastructure, and customer dissatisfaction. Studies conducted in various tourist places in Malaysia point out the gravity of this problem, for instance, Pulau Redang, Terengganu where the damage to the coral reefs has exceeded 80% because of over-tourism, Semporna, Sabah where there are problems of solid waste management and safety because of concentrated tourists, Cameron Highlands, Pahang where there are landslides and soil erosion caused by the tourism development in the area, and George Town, Penang where there is the UNESCO World Heritage place that is affected by concentrated tourists. Secondly, the less-visited states do not make use of tourism to spur local economies and hence lose out in terms of job creation. Though there are tourism data available, there is no systematic process to give an advance warning of the destinations that have reached their carrying capacities and also guide the dispersal of tourists to other destinations.

1.3 Objective

The purposes of this research project are to create a predictive model to forecast the number of domestic visitors to all 13 states and 3 federal territories of Malaysia for 2026 through time series analysis, to build a Tourism Carrying Capacity Index (CCI) for selected high-risk tourism destinations by considering economic, environmental, and infrastructural factors, to build an early warning dashboard showing capacity levels (green/yellow/red) per destination and provide recommendations for managers, and to develop a framework for decision-making that will allow stakeholders to predict the effect of their interventions.

2.Literature Review

2.1 Tourism Carrying Capacity

Tourism carrying capacity is defined as “the maximum number of tourists that can be accommodated before unacceptable negative effects start to occur”. Early attempts have been made to understand the concept of ecological carrying capacity. However, current studies reveal that carrying capacity is multidimensional, including economic, social, environmental and infrastructure carrying capacities.

In the Malaysian case, a carrying capacity index has been constructed on the island of Perhentian through Analytic Hierarchy Process (AHP). The weight given to the biophysical environment is 0.291, to planning control 0.25, to tourism facilities 0.229 and to socioeconomic 0.23. This will be a good base for the present study.

2.2 Over-Tourism and Its Consequences

Over-tourism happens where there is too much traffic that causes environmental destruction, infrastructure problems and the decline of the quality of life for the local population. In Malaysia, some of the destinations face the problem of over-tourism. For example, Redang Island in Terengganu, since excessive tourist activity (Reef Check Malaysia, 2025) the coral reef damage exceeded 80%. We discovered only 25.89% of coral cover in this area. Next, Tunku Abdul Rahman National Park located in Sabah and its coral cover is equal to 13.13% the smallest one in our research. Besides, Cameron Highlands in Pahang and its Landslides and soil erosion linked to tourism development with 2 landslide incidents recorded in 2025. In George Town, Penang the UNESCO World Heritage site faces pressure from concentrated tourist flows, with 800 tonnes of daily solid waste generated on Penang Island.

2.3 Tourism Demand Forecasting

Demand forecasting is crucial for destination management since it enables proper planning of resource distribution, infrastructure development, and the avoidance of overtourism.

Conventional techniques used in demand forecasting include Autoregressive Integrated Moving Average (ARIMA), Exponential Smoothing, and Artificial Neural Networks (ANN) (Song & Li, 2008). Nevertheless, these techniques do not take into account the problems of missing data, seasonality irregularities, and structural breaks inherent in Malaysian state-level tourism data.

The Prophet algorithm, created by Meta (previously called Facebook), can be considered an effective approach to overcoming these challenges. The algorithm involves the decomposition of time series into the components of trends, seasonality, and holidays, thus being quite explainable and resistant to outliers (Taylor & Letham, 2018). Comparison studies prove that Prophet outperforms other conventional models such as ARIMA and LSTM in forecasting the tourism dynamics, especially because it accounts for the non-linear nature of the processes, seasonality, and sudden shifts. Such characteristics make it particularly suitable for forecasting tourism in Malaysia, which is characterized by structural breaks (pandemic period of 2020-2021) in domestic tourism data and insufficient number of observations at the state level.

Considering the above features, Prophet becomes the selected forecasting model in this study. The forecasting model is estimated using annual domestic visitor numbers of all 13 states and 3 federal territories from 2018 to 2025, with the forecast extending to 2026 to 2028. The decomposition process of the model enables the incorporation of any structural break associated with the pandemic as a regressor, along with the uncertainty intervals of the forecast.

2.4 Multi-Criteria Decision Analysis in Tourism

Multiple criteria decision making (MCDA) techniques like AHP and Entropy Weight Method are popularly employed to develop composite indices. While AHP involves the expert's judgment, Entropy Weight Method is an alternative based on data-driven technique.

In our case, the weights obtained from AHP are taken from the Perhentian Island case study (Biophysical =0.291, Planning =0.250, Facilities = 0.229, Socioeconomic =0.230). The socioeconomic dimension is excluded because consistent destination-level data are not

available, and the remaining three dimensions are re-weighted to sum up to 1.0 (Biophysical = 0.378, Planning = 0.325, Facilities = 0.297).

2.5 Early-Warning Systems in Tourism

Systems of early warning (EWS) have emerged in disaster management, public health and finance. In the field of tourism, however, the EWS system has not been well-developed yet. The Global Sustainable Tourism Council (GSTC) stresses destination-level monitoring of sustainability on the dimensions of environment, society and economy.

An EWS for tourism is typically composed of three elements which is data gathering, indicators building and classification based on thresholds. This research follows this approach and uses predictive analytics (Prophet), capacity index (AHP-weighted CCI) and interactive dashboard.

2.6 Summary and Research Gap

While existing studies address tourism carrying capacity, over-tourism and demand forecasting, there is no integrated system in the Malaysian context that predicts future visitor flows at the state level, constructs a multidimensional carrying capacity index for high-risk destinations, visualises results in an interactive dashboard with decision-support features. This study fills this gap by developing a predictive analytics and smart destination redistribution system for sustainable tourism in Malaysia.

3. Methodology

3.1 Research Design

This study adopts a quantitative, data-driven approach to develop a tourism carrying capacity early-warning system for Malaysia. The methodology is structured into three analytical layers which is a predictive layer, forecasting domestic visitor flows for all 13 states and 3 federal territories from 2025 to 2028 using Prophet. Next, diagnostic layer, constructing a Tourism Carrying Capacity Index (CCI) for four high-risk destinations using AHP-weighted indicators. Last, decision support layer, clustering the 16 states using Hierarchical Clustering and visualising capacity levels in a green/ yellow/ red early warning dashboard.

3.2 Data Sources

Data were compiled from official Malaysian and International sources:

Indicator	Source
Domestic visitor arrivals (2019-2024)	DOSM (Department of Statistic Malaysia), Tourism Satellite Account
Hotel rooms, AOR, guest accommodation	Tourism Malaysia, DOSM State Tourism Reports
Coral cover	Reef Check Malaysia Annual Survey Report 2025
Non-Revenue Water (NRW)	SPAN (National Water Services Commision)
Non-domestic water consumption	OpenDOSM Water Consumption Catalogue
District population	OpenDOSM Population Catalogue
GDP by state	DOSM GDP by State 2025
Solid waste generation	Penang Island municipal reports (MBPP)
Landslide incidents	Pahang state government report (2025)
Visitor demand/ carrying capacity	Sabah Parks Dashboard, Terengganu state reports

Table 3.2.1 Data Source and Indicators

All raw data and source URLs are documented in the project's data repository (Null Datathon.xlsx), and a full list of indicators with their units, years, and sources is provided in Table 3.2.1.

3.3 Predictive Layer: Prophet Forecasting

The Prophet model (Taylor & Letham, 2018) is used to forecast annual domestic visitor arrivals for each of the 13 states and 3 federal territories for 2026. The specific models are the

training period from 2018 to 2025 which is 8 annual observations per state and its forecast horizon is 2026 (1 year). Its components are trending and yearly seasonality, with weekly and daily seasonality disabled due to the annual data frequency. Uncertainty, 80% prediction intervals are reported alongside the point forecast.

Prophet is chosen over ARIMA and LSTM due to the state level data having limited observation, the 2020-2021 pandemic creates a structural break that ARIMA cannot easily accommodate and Prophet handles missing data and irregular seasonality more robustly.

Master_data.xlsx already showed the data Growth2026_Forecast and Forecast_Uncertainty for each state.

3.4 Diagnostic Layer: Tourism Carrying Capacity Index (CCI)

The CCI is constructed for four high risk destinations which is Pulau Redang, Cameron Highlands, GeorgeTown and Tunku Abdul Rahman Park. Five indicators were used:

- I1: Non-Revenue Water (NRW), the proportion of treated water lost before reaching consumers, used as a proxy for water infrastructure efficiency.
- I2: Ecological Pressure, measured using coral cover for marine destinations, landslide incident count for Cameron Highlands, and river water quality index (WQI) for George Town.
- I3: Accommodation Pressure, based on the Average Occupancy Rate (AOR) in 2024.
- I4: Demand Growth Pressure, based on the 2026 forecast year-on-year growth rate.
- I5: Water Consumption, non-domestic water consumption in million litres per day (MLD).

Each indicator was normalised to a 0-1 scale, where higher values indicate greater stress. For Ecological Pressure (I2), data on coral cover was extracted from the Reef Check Malaysia Annual Survey Report 2025. Coral cover is measured as a ratio ranging from 0 to 1, with larger ratios denoting healthier reefs. Since larger values for coral cover denote smaller values for ecological pressure, the indicator was inverted and normalized based on the following formula:

$$S_{coral} = (coral_max - coral_cover) / (coral_max - coral_min)$$

Where coral_min= 0.1 and coral_max= 0.5 represented the minimum and maximum observed values of coral cover at the Malaysian coral reef location. Coral cover for Pulau Redang was found that 0.2589 (25.89%) with I2, Ecological Pressure value of 0.7411. For

Tunku Abdul Rahman Park, a coral cover of 0.1313 (13.13%) provides a score of 0.8687. As for Cameron Highlands, the number of landslides (2 landslides in 2025) was considered as the proxy and normalized with respect to the threshold value of 5. The River Basin Average WQI value of 74.125 for the George Town location was inverted and normalized. For those destinations, for which the particular proxy was not applicable, value 0.5 was assigned. AHP weight were derived from a pairwise comparison matrix adapted from the Perhentian Island carrying capacity study:

Indicator	AHP Weight
NRW	0.1452
Ecological Pressure	0.3704
Accommodation Pressure	0.2078
Demand Growth Pressure	0.0958
Water Consumption	0.1809

Table 3.3.1 AHP Weight

The consistency ratio (CR) of the AHP Matrix is 0.0167, which is far lower than the threshold of 0.1, thus indicating consistency. The EWM has been used for cross-validation of the results obtained, producing a correlation of 1.0000 between the EWM weights and the AHP weights.

The TCCI has been calculated by averaging the weighted and entropy weighted indices, and multiplying them by 1.5:

CCI Range	Status
$CCI < 0.7$	Green (Safe)
$0.7 \leq CCI < 1.00$	Yellow (Warning)
$CCI \geq 1.00$	Red (Critical)

Table 3.3.2 CCI Range

The scale factor of 1.5 was determined through calibration. Without scaling, all CCI values fall within a narrow range that would classify all destinations as Green under the 0.70

threshold. Scaling by 1.5 expands the distribution to align with the Green, Yellow, and Red thresholds.

Two validation approaches were used to assess the robustness of the CCI results. First, the consistency between AHP and EWM was evaluated using the Spearman rank correlation. Second, K-Means clustering with three clusters was applied to the scaled combined CCI scores to independently classify destinations into three groups. The cluster centroids were sorted and mapped to Green, Yellow, and Red labels based on their order. The rule-based classification and K-Means classification were then compared to assess the agreement rate.

3.5 Decision Support Layer: Hierarchical Clustering

The decision support layer was built subsequent to the diagnostic layer. This layer categorises the total of 13 states and 3 federal territories into different tourism pressure groups. It helps the decision makers prioritise and allocate resources for the destinations reaching or surpassing their carrying capacity.

Ward's method was used under hierarchical clustering because it minimises the within-cluster variance and is appropriate for small data sets with indicators in different scales. Unlike the K-Means, the hierarchical clustering does not necessitate the number of clusters to be decided prior to the analysis.

The clustering was performed based on 23 numeric features (see Table 3.2.1) from the master dataset (tourism data.xlsx) which includes information on tourism demand, accommodation, infrastructure, environment, and socio-economic dimension. Prior to the clustering process, z-score normalisation (StandardScaler) was applied to balance the impact of higher magnitude features (such as visitor arrivals) over the lower magnitude features (such as NRW ratios). Euclidean distance was used to calculate the pairwise distances and Ward's method to merge the clusters which minimise total within-cluster variance in each step.

Optimal number of clusters was determined from the dendrogram and domain knowledge.

The chosen three-cluster solution was “Low Pressure”, “Moderate Pressure”, and “High Pressure”. Clusters were labeled based on ranking of clusters according to the mean tourism pressure ratio (TOURISM_PRESSURE_2025).

Cluster outputs were tested with two validation techniques which are inspection of the dendrogram for confirmation of representativeness of the three-cluster solution and absence of over-segmentation, and heatmap of the means for confirmation of separation of the clusters and correctness of their labeling based on features.

The outputs of the clustering, including the dendrogram, assignment of clusters, and the heatmap of clusters' profile, are provided in Section 4.3.

3.6 Scenario Simulation (What-If Analysis)

Following the diagnostic and decision support layers, a what-if scenario simulation module was developed to evaluate the potential impact of different policy interventions on the Tourism Carrying Capacity Index (CCI) and on tourism revenue. This module allows policymakers to compare the trade-offs between reducing tourism pressure and maintaining tourism revenue before committing to a specific intervention.

Five scenarios were simulated: Baseline (no intervention), -20% visitors, -30% visitors, Infrastructure Investment (15% improvement in NRW and water efficiency), and Combined (-15% visitors + infrastructure).

For each of these scenarios, the following actions were undertaken.

Step 1 — Indicator adjustment. The relevant indicators were adjusted based on the scenario assumptions. For visitor reduction scenarios, the Demand Growth Pressure (I4) indicator was reduced by 60–75%, and the Accommodation Pressure (I3) indicator was reduced by 10–25%, reflecting the lower occupancy stress under reduced visitor volumes. For infrastructure investment scenarios, the NRW (I1) and Water Consumption (I5) indicators were reduced by 15%. The Ecological Pressure (I2) indicator was left unchanged across all scenarios, because ecological recovery is a slow process that cannot be achieved within a single planning cycle.

Step 2 — CCI recalculation. The adjusted indicators were multiplied by the AHP weights and scaled by the same factor (1.5) used in the baseline CCI computation (Section 3.4). This ensures that the scenario CCI scores are directly comparable to the baseline scores.

Step 3 – Estimate of revenues. Revenue estimate is obtained by multiplying the estimated number of visitors by average revenue per visitor. The value of average revenue per visit is derived from the DOSM Tourism Satellite Account 2025 that states RM 121.3 billion in expenditure on domestic tourism for 290.1 million domestic tourists, giving approximately RM 418 per trip. However, since the tourists visiting high-risk destinations spend more on hotels, diving and guided tours, a more conservative estimate of RM 2,500 per trip was taken as an upper bound for revenue impact analysis.

Step 4 – Classification of the Warning level based on the Recalculated CCI. Recalculated CCI was grouped according to the thresholds as follows; Green for any value below 0.70, Yellow from 0.70 to 1.00, and Red for any value equal to or above 1.00. This helps in identifying whether an intervention is sufficient to make the destination move from Yellow to Green or whether more drastic actions are needed.

Scenario Simulation Results are presented comprehensively in section 4.4. These results comprise the recalculated CCI, number of visitors, estimated revenues, difference in revenue compared to the baseline, and the warning level attained.

4. Finding

4.1 Predictive Layer: Domestic Visitor Forecasts (2026)

The Prophet model was trained on annual domestic visitor arrivals for all 13 states and 3 federal territories from 2018 to 2025, with the forecast horizon extended to 2026. The 2020–2021 pandemic period was incorporated as a structural break, and 80% prediction intervals were reported alongside point forecasts.

4.1.1 Forecast Results

The total predicted domestic visitors for 2026 is approximately 301.4 million, representing a continued recovery trend from the pandemic period. Table 4.1.1.1 presents the forecasted arrivals and year-on-year(YoY) growth rates for all 16 states and federal territories.

State	Predicted Visitors 2026(thousands)	YoY Growth 2026(%)
Selangor	39774	6.56
W.P Kuala Lumpur	31515	6.07
Perak	27603	8.25
Pahang	23754	6.89
Melaka	23544	8.57

Sabah	22631	5.92
Johor	21164	7.6
Negeri Sembilan	21123	7.26
Sarawak	20474	3.01
Pulau Pinang	18745	6.41
Kedah	17918	6.32
Terengganu	15854	5.38
Kelantan	12005	6.49
W.P. Putrajaya	2829	7.43
Perlis	3801	9.24
W.P. Labuan	361	-1.08

Table 4.1.1.1 Forecasted domestic visitor arrivals for 2026(Prophet Model)

Based on Table 4.1.1.1, Selangor recorded the highest predicted arrivals at 39774 thousand and 6.56% YoY growth, followed by W.P Kuala Lumpur with 31515 thousand and 6.07% YoY growth and Perak with 27603 thousand and 8.25%. These three states alone account for approximately 32.81% of total predicted arrival which is 301358 thousand. In contrast, W.P. Labuan recorded the lowest predicted arrivals at 361 thousand, followed by Perlis with 3801 thousand and W.P. Putrajaya with 2829 thousand.

4.1.2 Geographic Concentration

The distribution of predicted arrivals reveals a clear geographic concentration pattern. Selangor and W.P. Kuala Lumpur together account for approximately 23.7% of all predicted domestic visitors in 2026. The top five states are Selagor, W.P. Kuala Lumpur, Perak, Pahang and Melaka account for approximately 48.6% of total predicted arrivals. In contrast, the bottom five states which are W.P. Labuan, W.P. Putrajaya, Perlis, Kelantan and Terengganu account for only approximately 11.6% of total predicted arrivals. This imbalance reinforces the over-tourism risk identified in the Section 1.2 Problem Statement. High concentration

destinations face environmental and infrastructure stress, while less visited states remain under utilised, losing out on tourism driven economic development.

4.1.3 Implications for Demand Dispersal

Forecasting results serve as the quantitative justification of the decision-making layer introduced in Section 4.3 and indicate the states experiencing high tourist pressure and requiring appropriate interventions. States with high forecasts and high YoY arrivals, like Selangor, Melaka, and Perak, require demand management measures in the form of visitor limitations, price differentiation, and additional infrastructure. On the other hand, states with low numbers of tourists, like Perlis, W.P. Labuan, and W.P. Putrajaya would need promotion and infrastructure investments to bring more tourists.

4.2 Diagnostic Layer: Tourism Carrying Capacity Index(CCI)

4.2.1 Indicator Matrix

The normalised indicator matrix for the four high risk destinations is shown in Table 4.2.1.1. Higher values indicate greater stress.

Indicator Matrix:	I1_NRW	I2_Ecological_Pressure	I3_Accommodation_Pressure	I4_Demand_Growth_Pressure	I5_Water_Consumption
Pulau Redang	0.420	0.74110	0.375556	0.360000	0.243333
Cameron Highlands	0.485	0.40000	0.328889	0.460000	0.610000
George Town	0.284	0.25875	0.562222	0.426667	0.780000
Tunku Abdul Rahman Park	0.596	0.86870	0.386667	0.393333	0.460000
EWM weights: [0.14098791 0.42459028 0.09285729 0.01764886 0.32391566]					

Table 4.2.1.1 Normalised indicator matrix

The Ecological Pressure(I2) shows that Tunku Abdul Rahman Park has the highest ecological stress which is 0.8687, driven by its very low coral cover which is 13.13%. Pulau Redang follows with 0.7411 and its coral cover 25.89%. Cameron Highlands with 0.4 and George Town 0.2588 show lower ecological stress since they use landslide count and WQI as proxies.

4.2.2 AHP and EWM Weights

The AHP weights, derived from the pairwise comparison matrix, were:

Weight Comparison:			
Indicator	AHP	Entropy	Diff
I1_NRW	0.1452	0.1409	0.0043
I2_Ecological	0.3704	0.4244	0.0541
I3_Accommodation	0.2078	0.0928	0.1149
I4_Demand	0.0958	0.0180	0.0778
I5_Water	0.1809	0.3238	0.1429

The consistency ratio (CR) of the AHP matrix was 0.0167, well below the 0.1 threshold, confirming internal consistency.

The Spearman rank correlation between the AHP-weighted and EWM-weighted CCI scores was 1.0000($p=0.0000$) indicating perfect rank consistency between the two weighting approaches. Although the absolute weights differ, the relative ranking of destinations remains identical.

4.2.3 CCI Results

The CCI results, computed as the average of AHP-weighted and entropy-weighted indices and scaled by 1.5 are shown in Table 4.2.3.1 and Table 4.2.3.2

Capacity Index Results:						
Destination	AHP	Entropy	Combined	Scaled_AHP	Scaled_Ent	Scaled_Comb
Pulau Redang	0.4918	0.4938	0.4928	0.7378	0.7408	0.7393
Cameron Highlands	0.4413	0.4744	0.4579	0.6619	0.7117	0.6868
George Town	0.4359	0.4623	0.4491	0.6538	0.6934	0.6736
Tunku Abdul Rahman Park	0.6096	0.6446	0.6271	0.9144	0.9670	0.9407

Table 4.2.3.1 CCI results

Classification Results (using scaled combined TCCI):

Destination	Score	Level
● Pulau Redang	0.739	Yellow (Warning)
● Cameron Highlands	0.687	Green (Safe)
● George Town	0.674	Green (Safe)
● Tunku Abdul Rahman Park	0.941	Yellow (Warning)

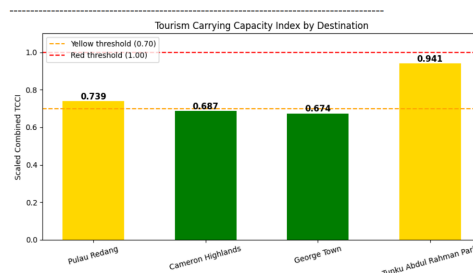


Table 4.2.3.2 classification results by destination

Both Pulau Redang and Tunku Abdul Rahman Park are categorized as “Yellow” (Warning), whereas Cameron Highlands and George Town are categorized as “Green” (Safe). The most worrying area among all the sites studied is the Tunku Abdul Rahman Park, whose total TCCI index stands at 0.941 due to extremely low coral coverage (13.13%), as well as relatively high NRW (0.596). Pulau Redang comes next with its scaled TCCI of 0.739 due to low coral coverage (25.89%) and medium demand pressure.

4.2.4 K-Means Validation

K-Means clustering ($k = 3$) was applied as an independent validation of the thresholds. The rule-based classification and K-Means clustering agreed on 3 out of 4 destinations (75% agreement).

Rule-based vs K-Means agreement: 75%

Destination	Rule-Based	K-Means	Match
Pulau Redang	Yellow (Warning)	Yellow (Warning)	✓
Cameron Highlands	Green (Safe)	Green (Safe)	✓
George Town	Green (Safe)	Green (Safe)	✓
Tunku Abdul Rahman Park	Yellow (Warning)	Red (Critical)	X

Table 4.2.4.1 K-Mean

4.2.5 Key Drivers of High CCI

The high CCI values for Pulau Redang and Tunku Abdul Rahman Park are driven primarily by ecological pressure (I2), reflecting coral cover below the 40% acceptable threshold used in the Perhentian Island study.

4.3 Decision Support Layer: TCCI-Based Early Warning System

4.3.1 Destination Risk Classification

The Tourism Carrying Capacity Index (TCCI) was translated into a three-level early warning system to support destination management decisions. Destinations with a TCCI below 0.70 were classified as **Green (Safe)**, values between 0.70 and 1.00 as **Yellow (Warning)**, and values of 1.00 or above as **Red (Critical)**. These categories provide a simplified interpretation of the composite index and allow tourism pressure to be communicated through the dashboard.

The results identify **Tunku Abdul Rahman Park as having the highest tourism carrying-capacity pressure**, with a TCCI of 0.941, followed by Pulau Redang at 0.739. Both destinations fall within the Yellow warning category. Cameron Highlands and George Town remain below the 0.70 threshold and are therefore classified as Green.

4.3.2 Early Warning Interpretation

The warning classification converts the numerical TCCI into information that can be interpreted more easily by tourism planners. A Green classification indicates that the destination remains below the model's warning threshold, while Yellow indicates that tourism pressure is approaching the predefined critical range and may require closer monitoring or intervention. Red represents the highest level of pressure under the proposed framework.

Importantly, a Green classification should **not be interpreted as the absence of tourism-related problems**. Instead, it indicates that the destination's combined indicators remain below the warning threshold established by this model. Among the four destinations, Tunku Abdul Rahman Park requires the greatest attention because its TCCI is closest to the Red threshold. Pulau Redang also enters the Yellow category, indicating elevated pressure relative to Cameron Highlands and George Town.

4.3.3 Decision-Support Application

The early warning system serves as the connection between the diagnostic analysis and management intervention. Rather than presenting the TCCI only as a numerical score, the Green–Yellow–Red classification allows decision-makers to quickly identify destinations that may require further attention. The system can therefore be interpreted as:

Tourism forecast → pressure indicators → TCCI → warning level → scenario evaluation → management response

For destinations classified as Green, the system supports continued monitoring of tourism and carrying-capacity indicators. Yellow destinations can be prioritised for scenario analysis to examine whether visitor-management or infrastructure interventions could reduce pressure. Destinations reaching Red would indicate that stronger management responses should be evaluated. This decision-support framework does not prescribe a single policy response. Instead, it provides a structured mechanism for identifying higher-pressure destinations and evaluating alternative interventions through the subsequent scenario simulation.

4.4 Scenario Simulation

4.4.1 Scenario Results

The scenario simulation evaluated five alternative management strategies for the four selected destinations: Baseline, 20% Visitor Reduction, 30% Visitor Reduction, Infrastructure Investment, and a Combined intervention involving a 15% visitor reduction with infrastructure improvement. The purpose was to assess whether tourism pressure could be reduced while considering the potential economic trade-off associated with each intervention.

Destination	Baseline	–20% Visitors	–30% Visitors	Infrastructu re	Combined
Pulau Redang	0.739	0.707	0.688	0.714	0.692
Cameron Highlands	0.687	0.630	0.610	0.621	0.598
George Town	0.674	0.612	0.585	0.613	0.583
Tunku Abdul Rahman Park	0.941	0.882	0.862	0.876	0.853

Table 4.4.1.1 Scenario Result

Table 4.4.1.1 shows that **Pulau Redang and Tunku Abdul Rahman Park were initially classified as Yellow (Warning)**, whereas Cameron Highlands and George Town remained within the Green (Safe) category. This is consistent with the baseline CCI results.

For Pulau Redang, a 20% reduction in visitors decreased the TCCI from 0.739 to 0.707 but was insufficient to change its warning category. A 30% reduction lowered the index to **0.688**, allowing the destination to move from Yellow to Green. The Combined scenario achieved a similar transition, reducing the TCCI to **0.692** with only a 15% reduction in visitors.

Tunku Abdul Rahman Park recorded the highest baseline TCCI at **0.941**. Although every intervention reduced its TCCI, none was sufficient to move the destination below the Green threshold of 0.70. The lowest value obtained was **0.853 under the Combined scenario**.

4.4.2 Trade-off Analysis

The simulation illustrates the trade-off between tourism-pressure reduction and estimated tourism revenue. Direct visitor restrictions produced larger reductions in tourism pressure but simultaneously reduced estimated visitor-related revenue. A 20% visitor restriction corresponded to a 20% decline in simulated revenue, while a 30% restriction corresponded to a 30% decline.

Infrastructure investment differs because it reduces the NRW and water-consumption pressure indicators without reducing visitor numbers. Consequently, the model produces a lower TCCI while maintaining the same simulated tourism revenue.

Pulau Redang illustrates this trade-off particularly clearly. The 30% visitor-reduction scenario moves the destination into the Green category but incurs the largest revenue reduction. In comparison, the Combined scenario also moves Pulau Redang into Green while limiting the simulated revenue reduction to **15%**.

For Tunku Abdul Rahman Park, however, even the strongest simulated interventions do not achieve Green status. This indicates that the interventions represented in the model are insufficient to reduce its tourism pressure below the predefined safety threshold.

4.4.3 Policy Implications

The scenario analysis indicates that a uniform tourism-management strategy may not be appropriate across destinations. Cameron Highlands and George Town already fall within the Green category under the baseline scenario; therefore, the simulation does not demonstrate a need for strong visitor restrictions solely to cross the CCI safety threshold.

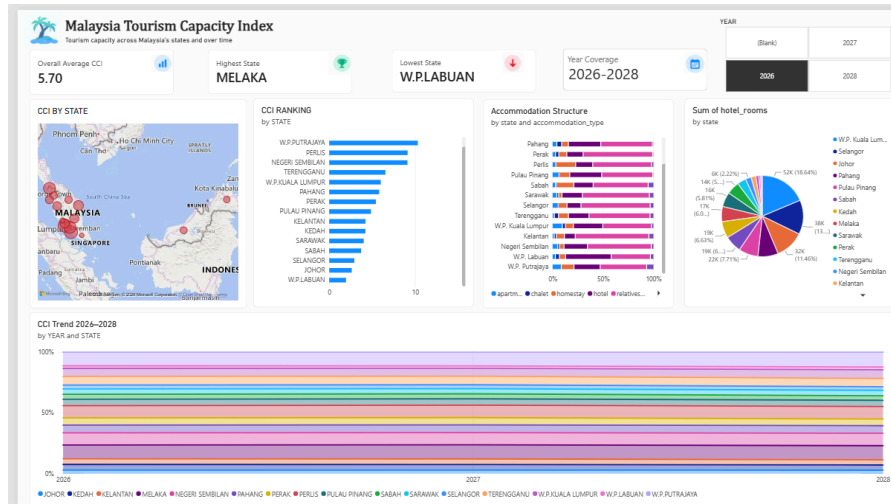
Pulau Redang presents a different situation. Both the 30% visitor-reduction scenario and the Combined scenario move the destination from Yellow to Green. The Combined scenario demonstrates that improvements in infrastructure together with a smaller visitor reduction can potentially reduce tourism pressure without requiring as large a reduction in tourism activity.

Tunku Abdul Rahman Park requires greater attention because it remains within the Yellow category under all five scenarios. This suggests that measures beyond the simulated short-term visitor and infrastructure interventions may be necessary to address its underlying carrying-capacity pressures.

Overall, the scenario simulation should be interpreted as a decision-support tool rather than a prescriptive policy model. It enables policymakers to compare how different intervention assumptions affect tourism pressure and the associated economic trade-offs.

5.Output

The final output of this study is an interactive two-page dashboard developed in Power BI, designed to translate the analytical results into an accessible decision-support tool for tourism managers and policymakers.



Dashboard Overview

The Malaysia Tourism Capacity Index dashboard provides an overview of tourism capacity across Malaysian states from 2026 to 2028. It presents key indicators, state-level comparisons, accommodation structures, hotel room distribution, and CCI trends over time. The dashboard includes a year slicer that allows users to switch between 2026, 2027, and 2028; all visualisations update automatically when a different year is selected. The dashboard provides four key indicators: the overall average CCI, the state with the highest CCI, the state with the lowest CCI, and the year coverage. For the selected year, the overall average CCI is 5.70, while Melaka has the highest CCI and W.P. Labuan has the lowest CCI.

CCI Comparison Across States

The CCI by State map provides a geographical overview of tourism capacity across Malaysia. The visualisation allows users to identify the distribution of CCI across different states and territories. When the slicer is set to 2026, Melaka shows the highest CCI. Switching to 2027 shows increased pressure for most states, while 2028 shows a slight easing but continued red status. The CCI ranking chart provides a more detailed comparison between states. Melaka records the highest CCI, followed by W.P. Putrajaya and Perlis, while W.P. Labuan records

the lowest CCI. The ranking makes it easier to identify states with relatively higher and lower tourism pressure.

Accommodation Structure

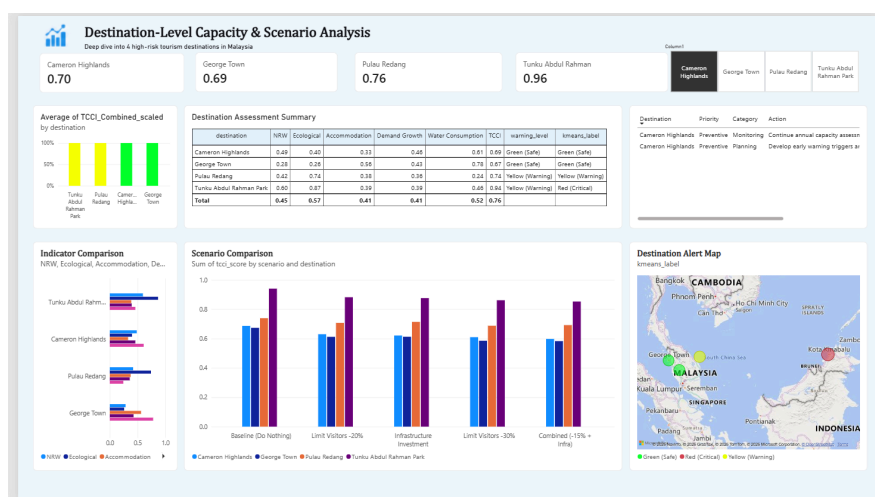
The accommodation structure chart illustrates the composition of different accommodation types across Malaysian states. The 100% stacked bar format allows the composition of accommodation types to be compared across states. The chart shows that relatives' and friends' houses dominate the accommodation structure (45–73% across states), while hotels account for only 10–45%. This indicates that most domestic tourists do not occupy commercial accommodation capacity, yet they still consume environmental resources such as water and waste services.

Hotel Room Distribution

The pie chart shows the distribution of hotel rooms across states. W.P. Kuala Lumpur has the largest share (52K rooms, 18.6%), followed by Selangor (38K, 13.6%) and Johor (32K, 11.5%). This confirms that commercial accommodation capacity is concentrated in urban and developed states.

CCI Trend Analysis

The CCI Trend chart illustrates the changes in tourism capacity across Malaysian states from 2026 to 2028. The CCI values show a slight upward trend from 2026 to 2027, followed by a minor decrease in 2028. Most states remain in the red zone throughout the period, suggesting that without intervention, tourism pressure will remain high.



Dashboard Overview

The dashboard focuses on four high-risk tourism destinations in Malaysia: Pulau Redang (Terengganu), Cameron Highlands (Pahang), George Town (Pulau Pinang), and Tunku Abdul Rahman Park (Sabah). While the first dashboard provides a state-level overview, this page provides a destination-level deep dive into the specific pressures facing each location.

Destination Capacity Index (KPI Cards)

Four KPI cards display the Tourism Composite Capacity Index (TCCI) for each destination. Tunku Abdul Rahman Park (TAR Park) has the highest TCCI at 0.96, followed by Pulau Redang (0.76), Cameron Highlands (0.70), and George Town (0.69). The TCCI values represent the 2026 baseline assessment for each destination. Unlike the state-level CCI, which is forecast for 2026–2028, the destination-level TCCI is calculated for a single year because the underlying indicators (coral cover, NRW, water consumption) are only available at annual resolution.

TCCI Ranking

A bar chart ranks the four destinations by TCCI from highest to lowest. TAR Park consistently ranks highest, indicating it is closest to overcapacity, while George Town ranks lowest. The ranking provides an immediate visual comparison of relative pressure across the four destinations.

Destination Assessment Summary

A detailed table lists the five indicators used to construct the TCCI: NRW (I1), Ecological Pressure (I2), Accommodation Pressure (I3), Demand Growth Pressure (I4), and Water Consumption (I5), together with the composite TCCI and warning level for each destination. Key findings show that TAR Park has the highest ecological pressure (0.87) and high NRW (0.60), indicating severe coral degradation and water infrastructure stress. Pulau Redang also has high ecological pressure (0.74), reflecting significant coral loss. George Town records the highest water consumption pressure (0.78), indicating high non-domestic water demand.

Cameron Highlands shows moderate pressure across all indicators, with the lowest ecological pressure.

Indicator Comparison

A grouped bar chart compares the five indicators across the four destinations, making it easy to identify which dimension drives pressure for each location. TAR Park scores highest on Ecological Pressure, while George Town scores highest on Water Consumption. This visualisation helps managers prioritise interventions based on the dominant pressure source.

Scenario Simulation (What-If Analysis)

A scenario comparison chart simulates five management strategies for each destination. Under the Baseline (Do Nothing) scenario, TCCI remains at its highest level. A 20% visitor reduction lowers TCCI but results in a 20% revenue loss, while a 30% reduction lowers TCCI further at the cost of a 30% revenue loss. Infrastructure investment also reduces TCCI, but without any revenue loss. The Combined scenario (–15% visitors + infrastructure) achieves the lowest TCCI with only a 15% revenue loss. The results show that infrastructure investment is more effective than visitor limits, as it reduces TCCI without reducing tourism revenue. For Pulau Redang, both the –30% visitor scenario and the Combined scenario move the destination from Yellow to Green. For TAR Park, however, none of the simulated interventions reduce the TCCI below the 0.70 threshold, suggesting that more drastic measures may be required.

Destination Alert Map

A map displays the four destinations coloured by K-Means cluster label. TAR Park with Red color means Critical. Pulau Redang with Yellow color means Warning. Cameron Highlands and George Town with green color means Safe. The map provides a geographical context for the destination-level analysis and highlights the spatial distribution of high-risk locations.

Recommendations Table

A recommendations table lists specific policy actions for each destination. TAR Park and Pulau Redang include real-time visitor counting, seasonal visitor management plans, and water infrastructure investment. Cameron Highlands and George Town include annual capacity assessment, early warning triggers, and response protocols. The table is filterable by destination, allowing policymakers to focus on the recommendations most relevant to their area.

6.0 Conclusion

A predictive and decision support framework for sustainable tourism was built in this research, which encompasses forecasting, carrying capacity assessment, and decision-support simulations in one package. This research attempts to fill a crucial need since there is no combined framework for sustainable tourism in Malaysia that forecasts future tourist visits, discovers destinations on the verge of exceeding their carrying capacities, and explores alternative management solutions.

The Prophet algorithm was employed for the predictive layer for forecasting future domestic tourist visits for all 13 states and 3 federal territories for the period of 2026-2028. Predictions for 2026 summed up to about 301.4 million visitors, yet the distribution stays very skewed with Selangor and W.P. Kuala Lumpur takes 23.7% of all tourists while the five least visited states account for 11.6%. The diagnostic layer included a Tourism Carrying Capacity Index (TCCI) calculation for four risky destinations with the help of five AHP-weighted indicators (NRW, Ecological Pressure, Accommodation Pressure, Demand Growth Pressure, Water Consumption). The consistency ratio of the AHP matrix is equal to 0.0167, and cross-validation of results of AHP and the Entropy Weight Method resulted in the Spearman correlation of 1.0000. K-Means clustering agreed with the rule-based classification 75% of the time. Tunku Abdul Rahman Park had the highest TCCI score (0.96), followed by Pulau Redang (0.76), Cameron Highlands (0.70), and George Town (0.69).

Five possible interventions were tested using the decision-support layer. An increase in infrastructure spending decreased TCCI, without impacting tourism income. In the case of Pulau Redang, both scenarios where there was a –30% reduction in the number of visitors, as well as Combined, turned Pulau Redang's color from Yellow to Green. None of the scenarios considered for Tunku Abdul Rahman Park decreased TCCI to below 0.70, meaning even stricter actions might need to be taken. Finally, the resulting output consists of an interactive two-page Power BI dashboard which implements all three analytical layers. Future work

should incorporate seasonal and monthly data, expand the indicator set to include social and economic dimensions, and validate the model with real-world intervention outcomes.

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